Technical Design: Multi-Source Candidate Data Transformer

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1. Pipeline / Step Breakdown

The system runs via a deterministic pipeline orchestrator:

1. **Ingest & Extract (Adapters)**: Fetch raw data concurrently from CSV, GitHub REST API, and Resume PDF. Each returns a RawRecord with a status (ok, failed, partial). Errors are caught and do not crash the pipeline.
2. **Normalize**: Pure functions standardize scalar values (phones to E.164, dates to YYYY-MM, countries to ISO-3166 alpha-2) and canonicalize skills using an explicit mapping.
3. **Fingerprint**: Generate a deterministic 16-character candidate_id via SHA-256 hash of sorted emails and phone numbers, with fallback to name hashing if absent.
4. **Merge & Resolve (Trust Strategy)**: Combines normalized records into a single CanonicalProfile. Source trusts (GitHub=0.85, PDF=0.75, CSV=0.70) dictate conflict resolution.
5. **Project to Output**: A runtime configuration reshapes the canonical schema dynamically.
6. **Validate**: A Zod schema verifies both the CanonicalProfile and the projected output.

2. Canonical Schema & Normalized Formats

- candidate_id: String (deterministic 16-char hex).
- full_name: String.
- emails: String Array (deduplicated, RFC-5322 regex validated).
- phones: String Array (deduplicated, **E.164** format via libphonenumber).
- location: { city, region, country } (country normalized to **ISO-3166 alpha-2**).
- links: { linkedin, github, portfolio, other[] }.
- skills: Array of { name, confidence, sources[] } (canonical names like javascript, machine-learning).
- experience: Array of { company, title, start, end, summary } (dates normalized to **YYYY-MM**).
- education: Array of { institution, degree, field, end_year }.
- provenance: Array of { field, source, method, raw_value? } tracking the origin of every field.
- overall_confidence: Number (0.0 to 1.0).

3. Merge Strategy & Conflict Resolution

- **Scalar Fields (Name, Headline, Location)**: Winner is the non-null value from the highest-trust source. Losing values are appended to provenance with method: 'discarded_conflict'.
- **Array Fields (Emails, Phones)**: Union across sources, deduplicated post-normalization.
- **Skills**: Unioned by canonical name. Confidence is the average trust of mentioning sources. If >= 2 sources agree, an agreement bonus (1.15x) is applied.
- **Experience**: The PDF acts as the primary source for rich prose. The CSV fills gaps if PDF experience is absent. Overlapping concurrent roles are kept but flagged in provenance.
- **Confidence Scoring**: field_confidence = source_trust × completeness_score × agreement_multiplier. Overall confidence is a weighted mean (identity and experience weighted 2x, others 1x). If no contact info exists, overall confidence is capped at 0.40 architecturally.

4. Runtime Custom-Output Config

The project() layer operates strictly on the internal canonical profile, avoiding any engine mutation. The configuration defines an array of field mappings:

- path: Output key name.
- from: Dot-bracket notation (e.g., emails[0]) resolved safely using a safeGet utility.
- normalize: Applies projection-level normalization overrides (e.g., forcing E.164 formatting on output).
- on_missing: Dictates behavior when a value is absent (null, omit, or error).
  The output is then separately validated against a dynamically generated schema constraint to ensure type compliance.

5. Input / Output Surface

The system provides two interfaces:

1. **Command-Line Interface (CLI)**: A robust CLI that processes files locally, fully supporting the custom runtime output configurations.
2. **Web UI (Vite + React)**: A complete graphical interface that runs concurrently with an Express API backend, allowing recruiters to upload CSVs and PDFs visually and instantly view the generated deterministic profiles.

6. Edge Cases

- **Missing / Malformed Source**: Handled gracefully. If GitHub rate limits (404/403) or a CSV row is completely empty, the adapter returns source_status: 'failed'. The pipeline continues merging remaining sources.
- **Ambiguous Phone Numbers**: A number like 9876543210 without a country code is ambiguous. If the candidate has a location with an identified country (e.g., IN), that is used as a hint for E.164 normalization. If no hint exists, the number is rejected rather than guessing a country code.
- **Unparseable Resume PDF**: The PDF adapter features a robust 3-stage fallback strategy. It first attempts pdf-parse; if that fails, it decodes raw compressed text streams; if the PDF is purely image-based, it dynamically falls back to native Tesseract OCR using a proportional aspect ratio.
- **Concurrent Roles**: Overlapping dates in experience (e.g., consulting while employed). The merge engine flags overlaps >3 months in provenance rather than overwriting them, preserving historical accuracy.
- **Descoped due to time**: Deep generative NLP extraction of unstructured PDF prose. Instead, robust regex heuristics are used for sectional boundary detection and line parsing to favor speed and determinism. Probabilistic name matching is also descoped in favor of strict SHA-256 fingerprinting for deterministic explainability.